

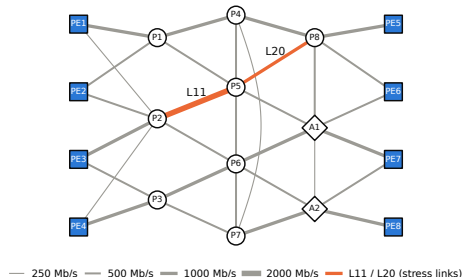
# Persistent state is not enough

A controlled study of planning horizons in incremental MPLS traffic engineering

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# 1. Problem: incremental MPLS traffic engineering

- ▶ 18 routers, 64 directed links, 17 demands with SLAs, 4 candidate LSPs each
- ▶ every 5 min: move *one* demand to another candidate, or do nothing
- ▶ moves cost reconfiguration effort; 3-interval hold-down after a move
- ▶ action mask: failed links, hold-down, protected-class capacity
- ▶ reward: delivery, disconnection, SLA severity, utilization, overload, minus move costs



## 2. Why RL — and why sequentiality should not be assumed

- ▶ routes **persist**, moves lock a demand, protected-class legality couples demands  $\Rightarrow$  formally an MDP
- ▶ but a move carries traffic *in the interval it is made*, traffic is exogenous
- ▶ recent TE systems treat TE as one-shot (Teal, DOTE) — in a setting *without* persistent state
- ▶ **Question:** here, does a learner that plans ( $\gamma = 0.995$ ) beat one that only predicts the immediate reward ( $\gamma = 0$ )? Why?

### 3. The surprising observation, reproduced

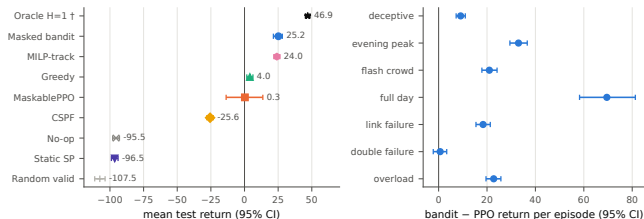
- ▶ closed predecessor study: bandit 18.2 vs PPO 9.0 (holdout return, 3 roots)
- ▶ retrained everything on different hardware, byte-identical training traffic:
  - ▶ baselines: **exact** ( $3 \cdot 10^{-14}$ )
  - ▶ bandit: **reproduces** (18.0)
  - ▶ PPO: **does not** (-12.0); per-root shifts up to 41 points
  - ▶ “PPO wins the deceptive scenario”: does not reproduce

| Learner | Root   | Selected (h./r.) | Hist. | Repr. |
|---------|--------|------------------|-------|-------|
| Bandit  | 42     | 250k / 400k      | 16.1  | 14.7  |
| Bandit  | 271828 | 400k / 350k      | 18.0  | 22.2  |
| Bandit  | 314159 | 300k / 250k      | 20.5  | 17.2  |
| PPO     | 42     | 250k / 350k      | 6.6   | -20.1 |
| PPO     | 271828 | 150k / 400k      | 12.2  | -29.0 |
| PPO     | 314159 | 350k / 200k      | 8.3   | 13.0  |

## 4. Methodology

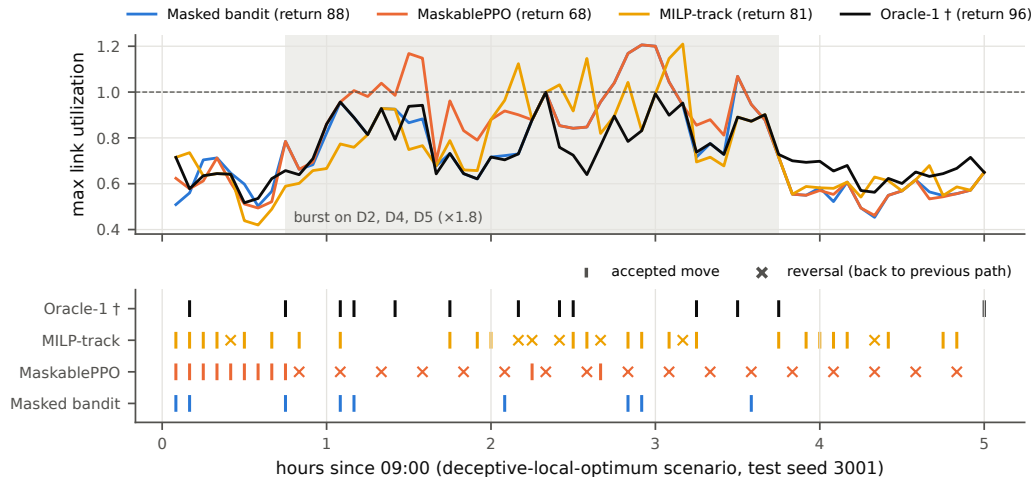
- ▶ same observation (604), mask, budget (400k transitions), paired traffic
- ▶ select on validation seeds, report on 140 disjoint test episodes; 5 training roots
- ▶ two-stage bootstrap CIs over roots and episodes
- ▶ references: static, greedy, CSPF, **MILP-track** (per-interval min-max-util), clairvoyant Oracle- $H$
- ▶ controls: PPO  $\gamma \in \{0, 0.9, 0.995\}$ ; masked Q-learner  $\gamma \in \{0, 0.5, 0.9, 0.99\}$  ( $\gamma = 0$  = bandit, bit-identical); PPO tuning; delayed-effect variant; time-of-day observation

## 5. Result: the bandit beats PPO and reaches the optimizer's return



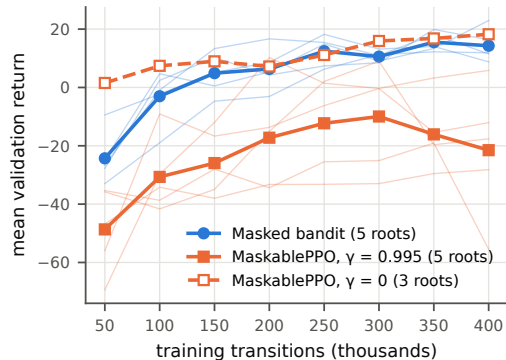
- ▶ bandit - PPO = 24.9,  $t$ -interval over roots [3.4, 46.4], positive on 5/5 roots
- ▶ bandit - MILP-track = 1.2 [-4.0, 6.3]; MILP has higher utility, more churn
- ▶ reroutes/h: bandit 2.30, MILP 5.83, PPO 6.38

## 6. Mechanism: PPO flaps at hold-down expiry



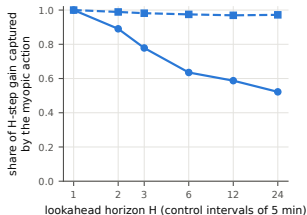
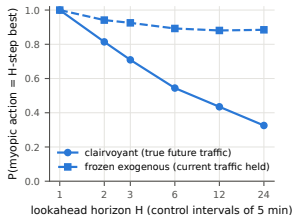
Utility: bandit 28.0, PPO 18.4; move costs: bandit -2.8, PPO -18.1

## 7. Attribution: the horizon, not the algorithm (so far)



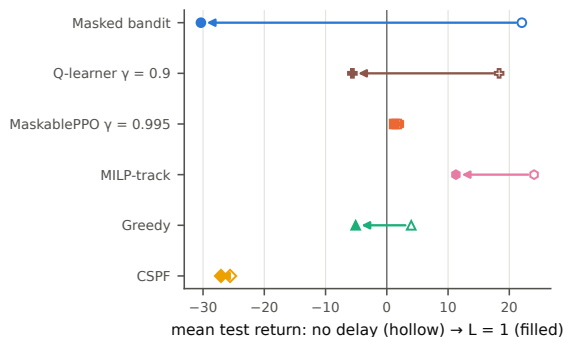
- ▶ PPO with  $\gamma = 0$ : **23.6** vs bandit 24.2 on the same roots
- ▶ difference -0.6 [-5.2, 2.7]
- ▶ flapping disappears
- ▶ small one-off costs are drowned by long-horizon advantage noise

## 8. How sequential is the problem?



- ▶ re-planning with perfect foresight: 3-step lookahead adds only 2.9
- ▶ open loop, true future: myopic = 24-step best in 33 % of states
- ▶ traffic frozen: **88 %**
- ▶ open-loop non-myopic value  $\approx$  anticipating exogenous change, not provided by the observation

## 9. When does the myopic learner fail? Delay the effect of a move



- ▶ move requested at  $t$ , active from  $t + 1$ : the immediate reward contains only its cost
- ▶ bandit: 22.0 → -30.3
- ▶ PPO: 1.6 → 1.2 (mean; per-root shifts in both directions)
- ▶ PPO — bandit under delay: 31.5 (mean), ahead on 2/2 roots
- ▶ MILP-track: 24.0 → 11.3 — still ahead of every learner
- ▶ shows: decision-time regression fails; not that long-horizon RL is needed

## 10. Clairvoyance: how much is lookahead worth?

- ▶ Oracle- $H$ : enumerates every legal move with the true future, no-op continuation
- ▶ Oracle-1 48.6, Oracle-3 51.4, Oracle-6 50.0 (5 test seeds)
- ▶ Oracle-3  $-$  Oracle-1 = 2.9 [1.5, 4.2]; Oracle-6  $-$  Oracle-1 = 1.4 [0.5, 2.5]
- ▶ bandit  $-$  Oracle-1 = -21.7: **knowing the next interval** is worth far more than planning beyond it

## 11. Limitations

- ▶ flow-level model, exogenous inelastic traffic, instantaneous control plane
- ▶ one topology, scripted scenarios, one reward; 5 (main) / 2–3 (ablation) roots
- ▶ single-move-per-interval formulation is a design choice

## 12. Take-aways and future work

- ▶ measure the value of lookahead (closed loop) before choosing a sequential learner
- ▶ a myopic learner (or a per-interval optimizer) is the baseline to beat
- ▶ long horizons can hurt by hiding small immediate costs
- ▶ future: delayed actuation and elastic traffic; packet-level validation; GNN policies; larger topologies; real traffic matrices